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AI-Driven Predictive Maintenance for Rainwater Harvesting Systems

CSA0925 – Programming in Java

Guided By:
Sivasankar Chandrasekran

Presented By:
Venkat Praveen (192373013)
M Krithick Rosan(192524206)
Prajaan K (192524380)

INTRODUCTION

- Rainwater harvesting is an environmentally friendly technique that collects and stores rainwater for domestic, agricultural, and industrial purposes, helping to reduce dependence on groundwater and municipal water supplies.
- The efficiency of a rainwater harvesting system depends on the proper functioning of components such as storage tanks, filters, pipelines, and water quality control units, all of which require regular maintenance.
- Traditional maintenance methods are based on manual inspections and periodic servicing, making it difficult to identify hidden faults such as clogged filters, pipe leakages, tank overflow risks, or poor water quality before they become serious problems.

OBJECTIVES

- Develop an intelligent Java-based predictive maintenance system that continuously monitors rainwater harvesting components and identifies potential maintenance issues before they cause system failures or reduce water collection efficiency.
- Implement AI-inspired rule-based analysis to evaluate sensor parameters such as water level, filter condition, pipe pressure, rainfall intensity, and water quality for accurate fault prediction and maintenance planning.
- Provide a user-friendly graphical dashboard that displays real-time system status, maintenance alerts, system health, and historical maintenance records for effective monitoring and decision-making.
- Promote sustainable water resource management by improving the reliability, efficiency, and lifespan of rainwater harvesting systems through intelligent predictive maintenance techniques.

PROBLEM STATEMENT

- Rainwater harvesting systems often experience maintenance issues such as clogged filters, overflowing storage tanks, leaking pipelines, and declining water quality, which significantly reduce overall system performance.
- Traditional maintenance methods rely on periodic manual inspections that consume considerable time, require human effort, and frequently fail to detect faults before major damage occurs.
- The absence of an intelligent monitoring mechanism leads to delayed fault detection, increased maintenance expenses, unnecessary water loss, and reduced efficiency in rainwater collection and storage.
- An automated predictive maintenance solution is required to continuously analyze system conditions, identify abnormal behaviour, and notify users about maintenance requirements

MODULE 1 :SENSOR DATA COLLECTION & MONITORING

Purpose

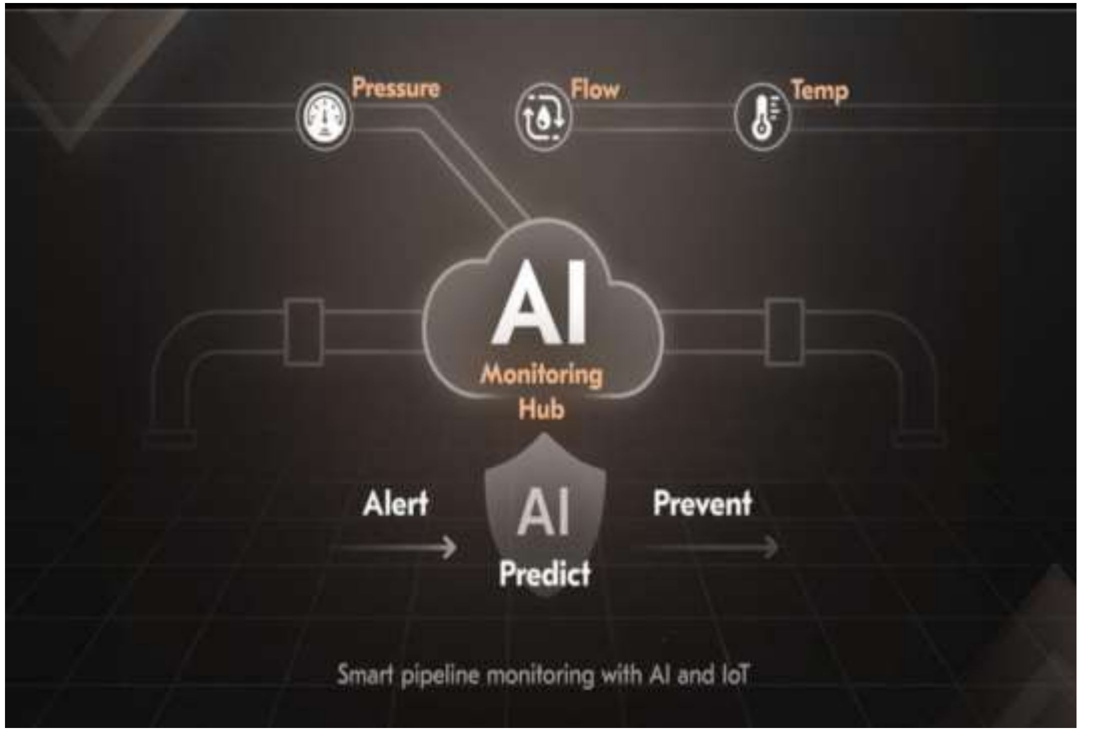
- To continuously collect, monitor, and validate operational data from different components of the rainwater harvesting system, ensuring that accurate information is available for intelligent fault prediction and maintenance analysis.

Functions

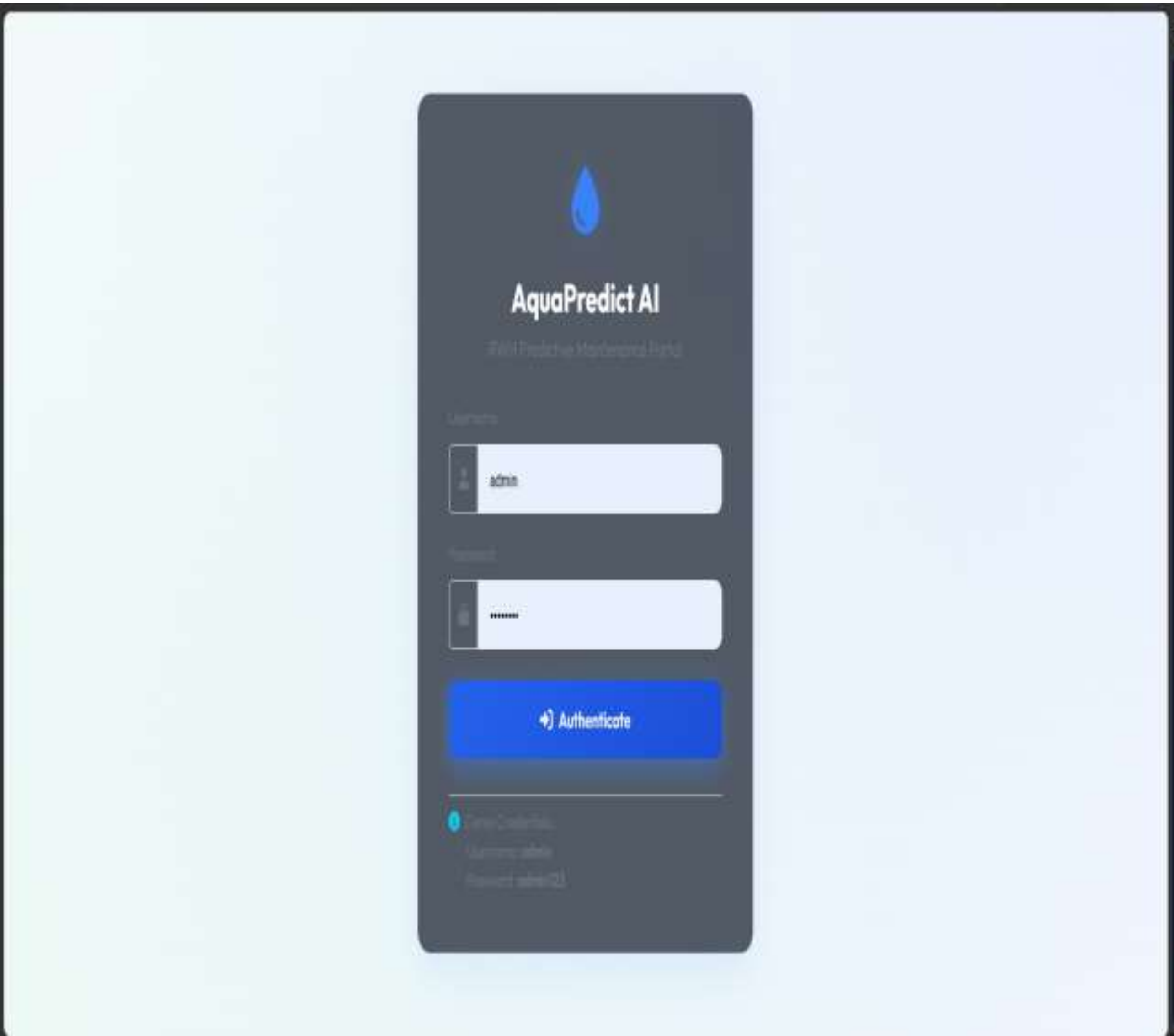
- Monitor water level inside the storage tank to identify low, normal, or overflow conditions.
- Collect filter condition data by measuring the amount dirt.
- Monitor water quality parameters to ensure the stored rainwater remains safe for future usage.

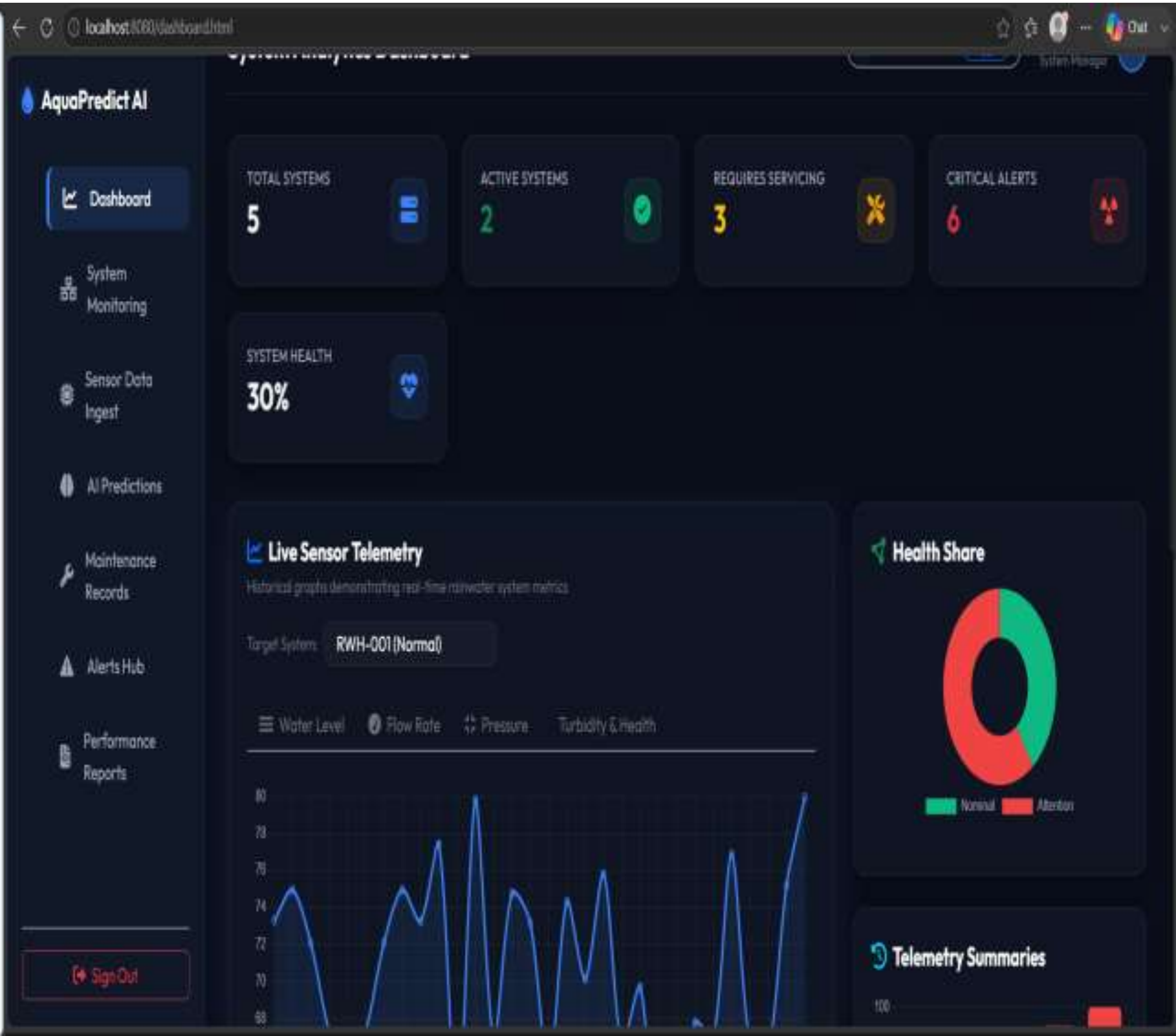
Working

- The monitoring module receives sensor readings from different parts of the rainwater harvesting system or accepts simulated.
- Each sensor value is validated to eliminate incorrect or abnormal input before processing.
- The validated sensor data is transferred to the AI Prediction Engine for intelligent fault detection.
- Continuous monitoring enables the application to maintain updated information about the overall system condition.



MODULE 1: IMPLEMENTATION





MODULE 2:AI PREDICTION ENGINE

Purpose

- To intelligently analyze the collected sensor information using AI-inspired rule-based algorithms and predict maintenance requirements before actual equipment failure occurs.

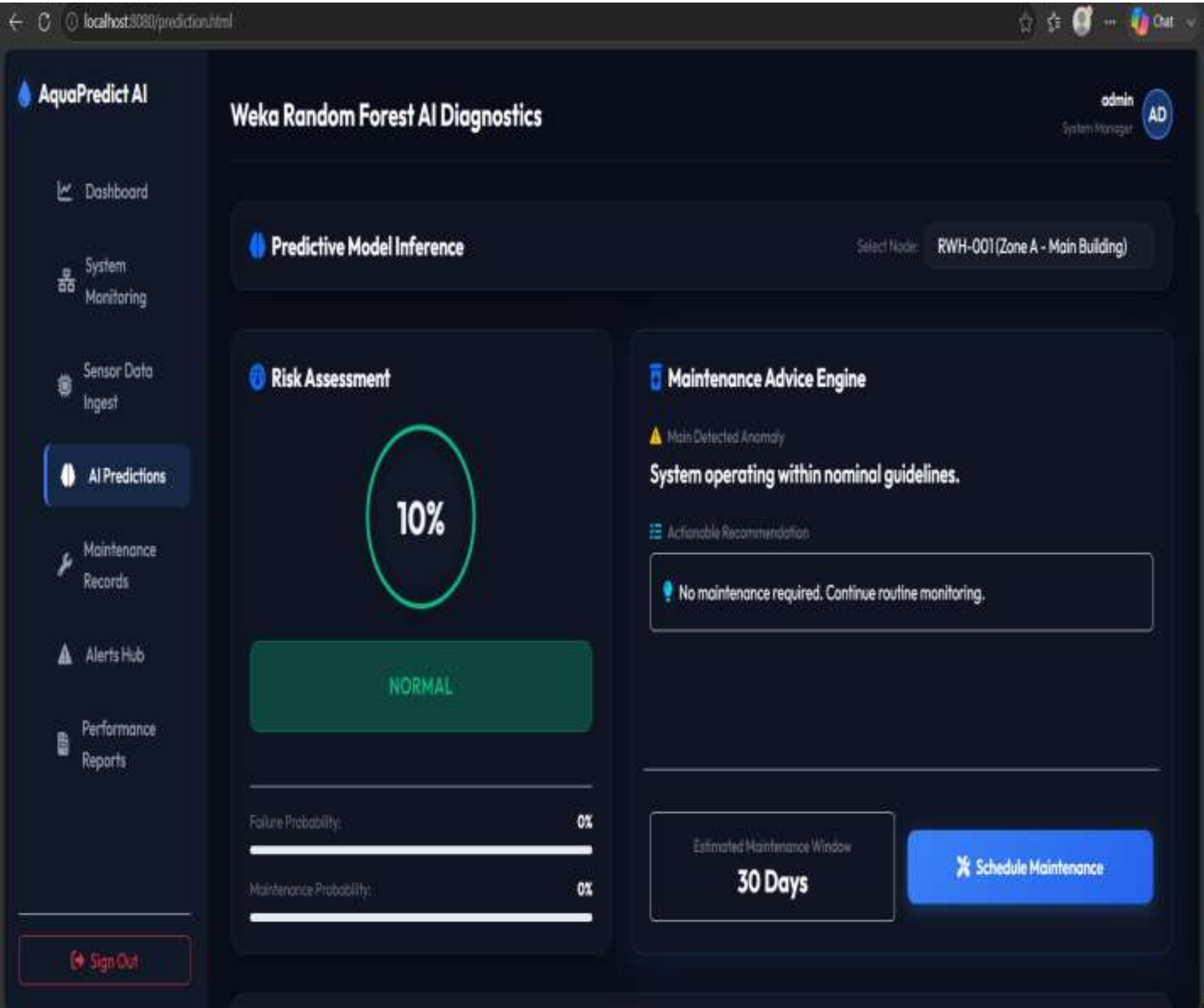
Functions

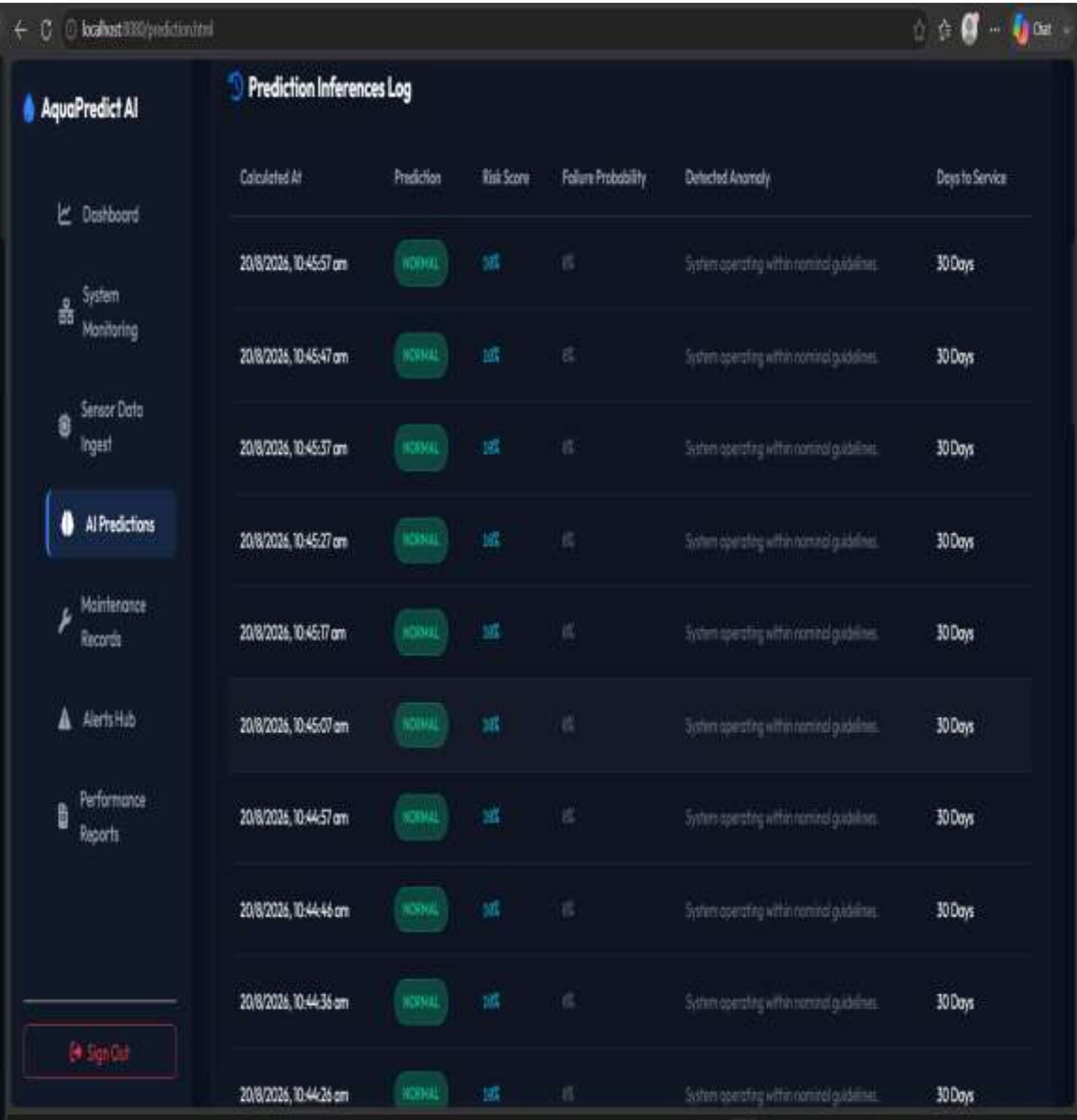
- Analyze water level, filter condition, pressure, rainfall, and water quality simultaneously.
- Compare current sensor readings with predefined threshold values.
- Detect abnormal operating conditions affecting system performance.
- Generate warning notifications whenever potential failures are identified.

Working

- The AI engine receives validated sensor data from the monitoring module.
- Each parameter is compared against predefined maintenance rules created within the Java application.
- The prediction results are forwarded to the maintenance recommendation module for further action.

MODULE 2 - IMPLEMENTATION





MODULE 3:MAINTENANCE RECOMMENDATION & REPORTING

Purpose

- To provide intelligent maintenance recommendations based on predicted faults, maintain maintenance history, and display overall system health through an interactive dashboard.

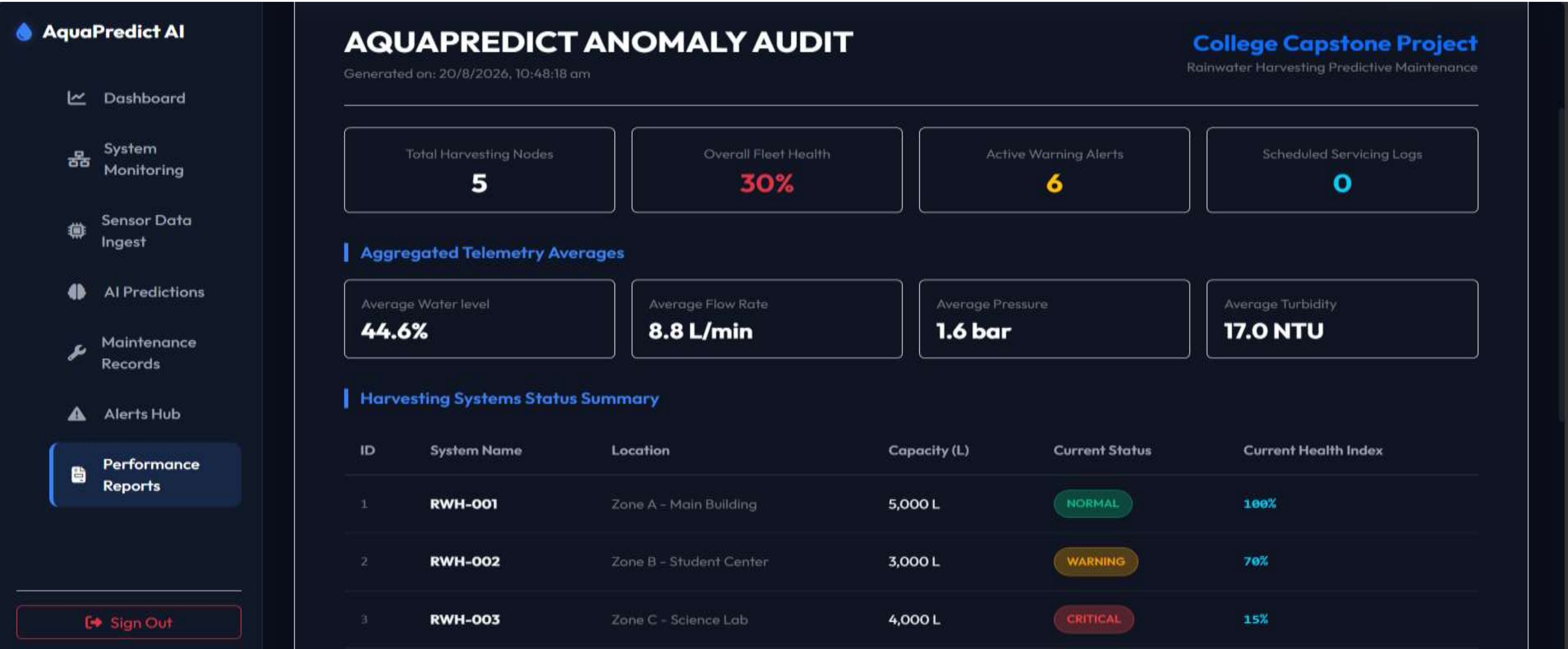
Functions

- Generate preventive maintenance recommendations.
- Display warning notifications for detected faults.
- Calculate overall system health score.
- Generate maintenance reports for users.

Working

- The module receives predicted fault information from the AI Prediction Engine.
- Maintenance alerts are displayed through the dashboard for immediate user attention.
- Maintenance records are saved using Java file handling for future monitoring and analysis.
- The dashboard continuously updates the overall health status of the rainwater harvesting system after every analysis.

MODULE 3 - IMPLEMENTATION



Future Scope

- Intelligent AI-inspired predictive maintenance using rule-based analysis.
- Continuous monitoring of rainwater harvesting system components.
- Automatic detection of filter clogging and overflow conditions.
- Early prediction of pipeline leakage and pressure abnormalities.
- Water quality monitoring for maintaining safe water storage.
- Real-time system health monitoring and reporting.
- Easy integration with future IoT-based smart water

Conclusion

- Developed a Java-based predictive maintenance system for rainwater harvesting.
- AI-inspired rule-based analysis predicts faults before system failure.
- Reduces water wastage and maintenance costs through early detection.
- Improves system efficiency, reliability, and sustainable water management.
- Future enhancements include IoT sensors, Machine Learning, and cloud integration.